

Rahul Kumar

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SKILLS

Languages:	C++, C, Python, JavaScript, TypeScript
Frontend:	React, Tailwind CSS
Backend:	Node.js, Express.js, RESTful APIs, JWT Authentication, Prisma, Drizzle ORM
Databases:	PostgreSQL, MongoDB
DevOps:	Docker, AWS (EC2), Terraform, Jenkins & GitHub Actions (CI/CD), Kubernetes, Prometheus, Grafana, Linux
Tools:	Git, GitHub, Postman, Redis, Jest, PyTest (Unit Testing & Test Automation)
AI-Assisted Dev:	Claude Code, GitHub Copilot, ChatGPT
Concepts:	Data Structures & Algorithms, OOP, System Design, Microservices, Distributed Systems

Projects

Aegis API (Distributed Microservices Gateway) Apr'26 – Jun'26

(Node.js, Express, PostgreSQL, Redis, Prisma, Docker) [GitHub](#)

- Architected a **4-service microservices backend** (API Gateway, Auth, User, Order) with Prisma/PostgreSQL, JWT auth (access + refresh tokens), and Redis-backed token blacklisting for stateful logout.
- Designed **role-aware distributed rate limiting** (Redis sliding window, 5–100 req/min by role) and per-service **circuit breakers (opossum)** with Prometheus-tracked state and a strict order-lifecycle state machine.
- Built full observability with **Prometheus + Grafana** dashboards and Winston structured logging; wrote **20 Jest/Supertest tests** reaching 78–93% coverage across services using ioredis-mock.
- Optimized Docker images via **multi-stage builds** (node:20-alpine), cutting image size from ~1.1GB to ~200MB.

Job Application Tracker (Flask, PostgreSQL, Docker, GitHub Actions) [Live Demo](#) Aug'2026

- Built and deployed a **multi-user job application tracking web app** (Flask, PostgreSQL, Docker) live on Render, with a ledger-style UI and **PyTest** unit tests in a **GitHub Actions CI/CD** pipeline.
- Used **AI-assisted coding tools (Claude Code, GitHub Copilot)** to accelerate debugging (Docker networking, import-order bugs, date-type mismatches) and speed up iteration.

URL Shortener API (Node.js, PostgreSQL, Drizzle, Docker, Jenkins, AWS) [GitHub](#) Feb'26 – Mar'26

- Built and deployed a production REST API (Node.js/Express, **Drizzle ORM**, PostgreSQL) to **AWS EC2**, with JWT auth, Zod request validation, and Helmet security headers.
- Designed a **Jenkins CI/CD pipeline** (install → test → Docker build → push to Docker Hub → SSH deploy) triggered on every push to main.
- Provisioned AWS infrastructure as code with **Terraform** (EC2 + security groups) and wrote **Kubernetes manifests** (Deployment, Service, HPA) for container orchestration.
- Instrumented the API with **Prometheus** and a 13-panel **Grafana** dashboard (latency percentiles, error rates, heap/GC metrics) for production monitoring.

Training

MERN Stack Developer | Lovely Professional University [Certificate](#) | Jul'25

- Completed hands-on training in MERN Stack Development – built scalable web apps using React.js, Node.js, Express.js, MongoDB.
- Built secure REST APIs with authentication, role-based access control, and CRUD/middleware patterns.
- Used GitHub for source control and collaborative development workflows.

Certificates

Cloud Computing NPTEL	Apr'26
Node.js- Beginner to Advance course with projects Udemy	Nov'25
Computer Networking Coursera	Dec'24
TCP/IP Advanced Protocols Coursera	Dec'24

Education

Lovely Professional University B.Tech – Computer Science and Engineering CGPA: 7.87	Phagwara, Punjab Aug'23 – Present
Kendriya Vidyalaya Intermediate (82%)	Khatkhati, Assam Apr'21 – Mar'22
King David School Matriculation (85%)	Khatkhati, Assam Apr'19 – Mar'20